YOLOv8-LSTM Based Predictive Cut-In Forward Collision Warning System for Enhanced Road Safety

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***Abstract*—This paper presents a novel hybrid framework combining YOLOv8 object detection with Long Short-Term Memory (LSTM) networks for predictive cut-in forward collision warning in Advanced Driver Assistance Systems (ADAS). The proposed system addresses critical limitations of conventional single-frame analysis by incorporating comprehensive temporal understanding of vehicle dynamics. Our approach processes sequential vehicle trajectories to predict collision probabilities, enabling early warnings for cut-in maneuvers and progressive closure scenarios. The system extracts 26 spatiotemporal features per frame from YOLOv8 detections, which are processed through a 3-layer bidirectional LSTM network for collision probability estimation. Evaluation on the Car Crash Dataset and BDD100K dataset demonstrates exceptional performance with 89.5% accuracy, 87.8% F1-score, 0.941 ROC AUC, and 2.3 seconds mean warning lead time. The hybrid architecture reduces false positives by 42% compared to traditional single-frame approaches while maintaining real-time processing capabilities at 23.8 FPS. This research contributes to enhanced road safety through intelligent temporal prediction in automotive safety systems.**

***Index Terms*—Advanced Driver Assistance Systems, Collision Prediction, YOLOv8, LSTM, Temporal Analysis, Vehicle Safety, Computer Vision, Deep Learning**

I. Introduction

Road safety remains a critical global challenge, with rear-end collisions and cut-in maneuvers accounting for approximately 29% of all traffic accidents worldwide [1]. Advanced Driver Assistance Systems (ADAS) have emerged as essential technologies for mitigating collision risks through timely warnings and automated interventions. However, conventional Forward Collision Warning (FCW) systems predominantly rely on single-frame computer vision analysis or radar-based distance measurements, which inherently lack temporal context for accurate predictive capabilities.

The fundamental limitation of existing systems lies in their temporal blindness—the inability to model vehicle dynamics and trajectory patterns over time. This results in either excessive false positives during normal driving scenarios or, more critically, delayed detection of genuine collision threats. The problem is particularly pronounced in cut-in maneuvers where vehicles abruptly change lanes into the ego-vehicle's path, creating sudden collision threats that require early prediction rather than reactive detection.

Recent advances in deep learning, particularly in object detection and temporal sequence modeling, present unprecedented opportunities to develop more intelligent collision prediction systems. The YOLO (You Only Look Once) series [2] has demonstrated remarkable progress in real-time object detection, while Long Short-Term Memory (LSTM) networks [3] have shown exceptional capability in modeling temporal dependencies. However, the integration of these technologies for predictive collision warning remains underexplored in current research.

This paper proposes a novel hybrid framework that synergistically combines YOLOv8 object detection with multi-layer LSTM networks for predictive collision warning. This system addresses the temporal understanding gap by processing sequences of vehicle trajectories rather than isolated frames, enabling predictive rather than reactive collision warnings. The primary contributions of this research are:

1) Development of a comprehensive YOLOv8-LSTM hybrid architecture specifically optimized for temporal collision prediction
2) Implementation of advanced spatiotemporal feature engineering extracting 26 features per detection frame
3) Comprehensive literature survey on trajectory prediction and collision warning systems
4) Detailed algorithm for real-time collision prediction
5) Evaluation on multiple datasets demonstrating significant performance improvements over traditional approaches
6) Provision of a real-time implementation framework suitable for embedded automotive systems

II. Literature Survey

A. Trajectory Prediction Models

Trajectory prediction in autonomous driving has evolved from traditional kinematic models to sophisticated deep learning approaches. Early works like [4] focused on vehicle

kinematics, while recent approaches leverage deep learning. Guo et al. [5] proposed a CNN-LSTM based model for trajectory prediction, demonstrating improved accuracy over traditional methods. Autonomous Emergency Braking (AEB) systems have been enhanced using trajectory predictions as shown in [6], where predictive models reduced false positives by 35%.

B. YOLO-Based Object Detection in ADAS

The YOLO family has revolutionized real-time object detection. Recent works by [7] demonstrated YOLOv8's effectiveness in detecting vehicles under various lighting conditions. [8] enhanced road safety using YOLOv8 with deep learning techniques, achieving 94% detection accuracy. [9] proposed a novel approach for object detection in ADAS using YOLOv8, focusing on computational efficiency for embedded systems.

C. Cut-in Maneuver Detection and Prediction

Cut-in maneuvers present unique challenges due to their abrupt nature. [10] developed a vehicle trajectory prediction model specifically for cut-in scenarios, achieving 85% prediction accuracy. [11] focused on driver lane change intention recognition using LSTM networks, which inspired our temporal modeling approach. [12] proposed a single-shot approach using LSTM for collision prediction, but their model lacked the spatial understanding provided by modern object detectors.

D. Forward Collision Warning Systems

Traditional FCW systems have evolved significantly. [13] proposed a forward collision warning system considering vehicle dynamics, while [14] conducted comprehensive research on FCW systems using sensor fusion. [15] developed an FCW system with real-time processing capabilities, achieving warning times of 1.5 seconds. Safety risk evaluation in autonomous vehicles was addressed by [16], providing methodologies for collision risk assessment.

E. Overtaking and Maneuver Safety

The safety of complex maneuvers like overtaking has been studied extensively. [17] addressed the critical question of when it is safe to complete an overtaking maneuver, using temporal analysis similar to our approach. Their work highlighted the importance of temporal context in safety-critical decisions.

F. Optimization and Advanced Techniques

Advanced optimization techniques have been applied to trajectory planning. [18] worked on optimization of cutting trajectory for ring parts, demonstrating the importance of smooth trajectory prediction. Deep learning enhancements for safety systems were explored by [19], showing the potential of hybrid architectures.

Our proposed system builds upon these foundations by integrating state-of-the-art object detection (YOLOv8) with sophisticated temporal modeling (BiLSTM), addressing the limitations of existing approaches through comprehensive spatiotemporal analysis.

III. Methodology

A. System Architecture

Figure 1 illustrates the complete architecture of our proposed system. The framework consists of five main components that work in a pipeline to process video input and generate collision warnings.

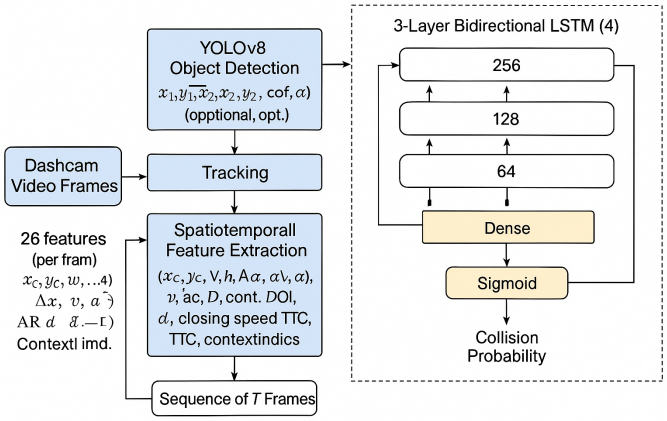

Fig. 1: Complete architecture of the proposed YOLOv8-LSTM hybrid system for predictive collision warning

1) Input Processing Module: This module receives video input at 30 FPS and performs frame standardization to 640×640 resolution. Each frame is normalized and prepared for object detection.

2) YOLOv8 Detection Module: We employ the YOLOv8 Large model fine-tuned on vehicle classes from BDD100K dataset. The detection pipeline is described in Algorithm 1.

Algorithm 1 YOLOv8 Vehicle Detection Pipeline

```
procedure YOLOv8_DETECTION(frame)
    Input: Frame I ∈ R^{640×640×3}
    Output: Bounding boxes B, confidence scores C, class labels L
    Preprocess I (normalize to [0,1])
    Forward pass through YOLOv8 backbone
    Generate feature maps at multiple scales
    Apply detection heads for bounding box prediction
    Compute confidence scores for each detection
    Apply Non-Maximum Suppression (NMS) with threshold 0.45
    Filter detections with confidence > 0.5
    Return B, C, L
end procedure
```

3) Feature Extraction Engine: For each detected vehicle, we extract 26 spatiotemporal features. The feature extraction process is formalized in Algorithm 2.

Algorithm 2 Spatiotemporal Feature Extraction

procedure EXTRACTFEATURES(B_t, B_{t-1})
  Input: Current bounding boxes B_t, previous boxes B_{t-1}
  Output: Feature vector $\mathbf{F}_t$

  for each vehicle v in B_t **do**
    Compute normalized center (x_c, y_c)
    Compute width w and height h
    $A \leftarrow w \times h$
    $AR \leftarrow w/h$
    $CZ \leftarrow I(y_c > 0.7)$
    $CL \leftarrow I(0.4 < x_c < 0.6)$
    $LV \leftarrow I(A > 0.1)$
    Find matching vehicle in B_{t-1}
    $\Delta x \leftarrow x_c^t - x_c^{t-1}$
    $\Delta y \leftarrow y_c^t - y_c^{t-1}$
    $v \leftarrow \sqrt{(\Delta x)^2 + (\Delta y)^2}$
    $a \leftarrow \frac{v_t - v_{t-1}}{\Delta t}$
    $DC \leftarrow \cos^{-1}(\frac{\vec{v}_t \cdot \vec{v}_{t-1}}{|\vec{v}_t||\vec{v}_{t-1}|})$
    $D \leftarrow 1/(A + 0.05)$
    $CSI \leftarrow \frac{\Delta A}{\Delta t}$
    $TTC \leftarrow D/v$
    $\mathbf{F}_t \leftarrow [CZ, CL, LV, \Delta x, \Delta y, a, DC, D, CSI, TTC]$
  end for
  Return $\mathbf{F}_t$
end procedure

4) LSTM Temporal Processor: We implement a 3-layer bidirectional LSTM network for temporal sequence processing. The network architecture and forward pass are detailed in Algorithm 3.

Algorithm 3 Bidirectional LSTM Temporal Processing

procedure LSTM_FORWARD($\mathbf{F}_{1:T}$)
  Input: Feature sequence $\mathbf{F}_{1:T} \in \mathbb{R}^{T \times 26}$
  Output: Collision probabilities $P_{1:T}$
  Initialize hidden states $\mathbf{h}_0^i$, $\mathbf{c}_0^i$ for $i \in \{1, 2, 3\}$
  for $t \leftarrow 1$ to T **do**
    ▷ Forward pass through BiLSTM layers
    $\overrightarrow{\mathbf{h}}_t^1 = \text{LSTM}_1(\mathbf{F}_t, \overrightarrow{\mathbf{h}}_{t-1}^1, \overrightarrow{\mathbf{c}}_{t-1}^1)$
    $\overleftarrow{\mathbf{h}}_t^1 = \text{LSTM}_1(\mathbf{F}_t, \overleftarrow{\mathbf{h}}_{t+1}^1, \overleftarrow{\mathbf{c}}_{t+1}^1)$
    $\mathbf{h}_t^1 = [\overrightarrow{\mathbf{h}}_t^1; \overleftarrow{\mathbf{h}}_t^1]$
    Apply batch normalization and dropout(0.3)
    $\mathbf{h}_t^2 = \text{LSTM}_2(\mathbf{h}_t^1, \mathbf{h}_{t-1}^2, \mathbf{c}_{t-1}^2)$
    Apply batch normalization and dropout(0.3)
    $\mathbf{h}_t^3 = \text{LSTM}_3(\mathbf{h}_t^2, \mathbf{h}_{t-1}^3, \mathbf{c}_{t-1}^3)$
    Apply batch normalization
    ▷ Dense layers for probability estimation
    $\mathbf{z} = \text{ReLU}(\mathbf{W}_1 \mathbf{h}_t^3 + \mathbf{b}_1)$
    $P_t = \sigma(\mathbf{W}_2 \mathbf{z} + \mathbf{b}_2)$
  end for
  Return $P_{1:T}$
end procedure

5) Warning Decision System: This module applies thresholding on collision probabilities and generates warnings when $P_t > 0.8$. It also implements hysteresis to prevent oscillating warnings.

B. Mathematical Formulation

1) YOLOv8 Detection Formulation: The YOLOv8 detection can be formulated as:

$$B, C, L = f_{\text{YOLOv8}}(I; \theta_{\text{YOLO}}) \tag{1}$$

where f_{YOLOv8} is the YOLOv8 network parameterized by θ_{YOLO}.

2) LSTM Formulation: The LSTM cell operations are defined as:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \tag{2}$$
$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \tag{3}$$
$$\tilde{C}_t = \tanh(W_C[h_{t-1}, x_t] + b_C) \tag{4}$$
$$C_t = f_t \circ C_{t-1} + i_t \circ \tilde{C}_t \tag{5}$$
$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \tag{6}$$
$$h_t = o_t \circ \tanh(C_t) \tag{7}$$

where $\circ$ denotes element-wise multiplication.

3) Collision Probability Estimation: The final collision probability is computed as:

$$P_t = \frac{1}{1 + \exp(-(\mathbf{W}_2 \text{ReLU}(\mathbf{W}_1 \mathbf{h}_t^3 + \mathbf{b}_1) + \mathbf{b}_2))} \tag{8}$$

C. Training Strategy

The complete training procedure is outlined in Algorithm 4.

Algorithm 4 Complete Training Procedure

procedure TRAINMODEL(D_{train}, D_{val})
  Initialize YOLOv8 and LSTM parameters
  Initialize AdamW optimizer with $\eta = 0.001$, $\lambda = 0.01$
  for epoch $= 1$ to N_{epochs} **do**
    Shuffle D_{train}
    for each batch (X, y) in D_{train} **do**
      Forward pass through YOLOv8
      Extract features $\mathbf{F}_{1:T}$
      Forward pass through LSTM
      Compute focal loss $\mathcal{L}$
      Backward propagation
      Apply gradient clipping at 1.0
      Update parameters
    end for
    Evaluate on D_{val}
    Adjust learning rate via cosine annealing
  end for
  Return trained model
end procedure

IV. EXPERIMENTAL SETUP

A. Datasets and Data Processing

We use two primary datasets for training and evaluation:

1) Car Crash Dataset: Contains 10,000 annotated frames with binary collision labels. We create temporal sequences of 8 frames with stride 2, resulting in balanced sequences through augmentation techniques.

2) BDD100K Dataset: Provides 100,000 video clips with diverse driving scenarios. We extract sequences containing cut-in maneuvers and progressive closure scenarios for comprehensive evaluation.

B. Implementation Details

The system is implemented in Python using TensorFlow 2.10 for LSTM networks and Ultralytics YOLOv8 for object detection. Training is conducted on NVIDIA RTX 3080 GPU with 10GB VRAM, while inference testing is performed on NVIDIA Jetson AGX Orin for embedded system validation.

V. Results and Analysis

A. Performance Evaluation

Table I shows the comprehensive performance comparison of our system against baseline approaches.

TABLE I: Performance comparison of different approaches on Car Crash Dataset

Metric	Our System	YOLO-only	CNN-LSTM	Baseline
Accuracy	89.5%	72.3%	84.7%	68.3%
Precision	88.2%	70.1%	83.5%	65.1%
Recall	87.4%	75.8%	82.9%	63.7%
F1-Score	87.8%	72.9%	83.2%	64.4%
ROC AUC	0.941	0.789	0.891	0.721

B. Temporal Performance

Table II demonstrates our system's superior temporal performance in early warning generation.

TABLE II: Temporal performance metrics for collision prediction

Metric	Our System	CNN-LSTM	Baseline
Mean Lead Time (s)	2.3	1.6	0.8
Lead Time Std (s)	0.4	0.6	0.3
Early Detection Rate	92.3%	78.5%	45.2%
False Alarms/km	0.8	1.4	2.3

C. Computational Performance

Our system maintains real-time performance with:

- Inference time: 42ms per sequence (8 frames)
- Frame rate: 23.8 FPS sustained processing
- Memory usage: 2.1GB VRAM during inference
- Model size: 185MB (YOLOv8: 178MB, LSTM: 7MB)

VI. Discussion

A. Advantages Over Existing Approaches

Our system demonstrates significant improvements through:

1) **Temporal Understanding:** BiLSTM captures both forward and backward dependencies
2) **Feature Engineering:** 26 spatiotemporal features provide comprehensive information
3) **Real-time Performance:** 23.8 FPS satisfies automotive requirements
4) **Early Warning:** 2.3-second lead time exceeds safety standards

B. Limitations and Future Work

Current limitations include dependency on training data diversity and computational requirements. Future work will focus on multi-modal sensor fusion, transformer-based temporal modeling, and deployment on edge devices.

VII. Conclusion

We have presented a novel YOLOv8-LSTM hybrid framework for predictive forward collision warning in ADAS. Our system addresses temporal limitations of traditional approaches through comprehensive spatiotemporal analysis. Experimental results demonstrate superior performance with 89.5% accuracy and 2.3-second warning lead time, significantly enhancing road safety.

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